

Unit 5 Guided Notes — The Pythagorean Theorem

$a^2 + b^2 = c^2$. Add the legs to find the hypotenuse; subtract to find a leg.

Standards: NC.8.G.6 · NC.8.G.7 · NC.8.G.8

This packet has three parts: **Part 1 — Guided Notes**, **Part 2 — Practice Problems**, and **Part 3 — Answer Key**. Fill the blanks on your printed copy first, then click any blank, step, or check button to reveal the answer.

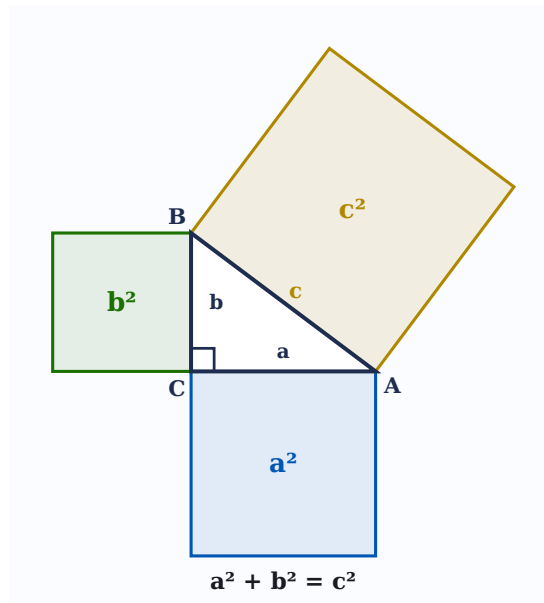
Topics Covered: 5.1 The Pythagorean Theorem — Parts of a Right Triangle (NC.8.G.6) · 5.2 Finding the Hypotenuse (NC.8.G.7) · 5.3 Finding a Missing Leg (NC.8.G.7) · 5.4 Converse of the Pythagorean Theorem (NC.8.G.6) · 5.5 Word Problems (NC.8.G.7) · 5.6 Distance on the Coordinate Plane (NC.8.G.8)

1 The Pythagorean Theorem

✦ **THE THEOREM:** In a RIGHT triangle, the sum of the squares of the two LEGS equals the square of the HYPOTENUSE.

$$a^2 + b^2 = c^2$$

a, b = legs (shorter sides) | c = hypotenuse (longest side, across from the right angle)



⚠ **IMPORTANT:** The Pythagorean Theorem ONLY works on RIGHT triangles (triangles with a 90° angle)!

Fill in the Blanks

Complete each blank on your printed copy, then click to check.

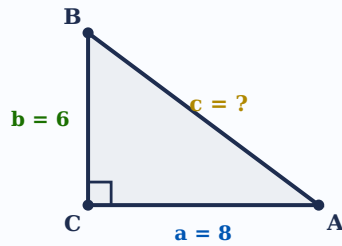
The two shorter sides of a right triangle are called the _____ (labeled a and b). The longest side, directly across from the right angle, is the _____ (labeled c).

The theorem written as an equation: $a^2 + b^2 =$ _____. It applies only to _____ triangles.

2 Finding the Hypotenuse

✦ **WHEN TO USE:** When you know BOTH LEGS and need to find the HYPOTENUSE. Formula: $c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$

FIND THE HYPOTENUSE



Example: Legs are 6 and 8. Find the hypotenuse.

Step 1 – Write formula: $a^2 + b^2 = c^2$

Step 2 – Substitute: $6^2 + 8^2 = c^2$

Step 3 – Square: $36 + 64 = c^2$

Step 4 – Add: $100 = c^2$

✓ Step 5 – Square root: $c = \sqrt{100} = 10$

Practice — Find the Hypotenuse (c)

CALC Legs 3 and 4: $3^2 + 4^2 = c^2 \rightarrow c = ?$

C =

CALC Legs 6 and 8: $6^2 + 8^2 = c^2 \rightarrow c = ?$

C =

CALC Legs 5 and 12: $5^2 + 12^2 = c^2 \rightarrow c = ?$

C =

CALC Legs 8 and 15: $8^2 + 15^2 = c^2 \rightarrow c = ?$

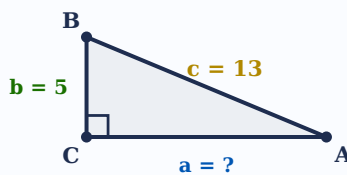
C =

3 Finding a Missing Leg

✦ **WHEN TO USE:** When you know ONE LEG and the HYPOTENUSE. Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{c^2 - b^2}$

💡 Finding a LEG = SUBTRACT from the hypotenuse squared. Finding the HYPOTENUSE = ADD the two legs squared.

FIND THE MISSING LEG



Example: One leg is 5, hypotenuse is 13. Find the other leg.

Step 1 – Write formula: $a^2 + b^2 = c^2$

Step 2 – Substitute: $5^2 + b^2 = 13^2$

Step 3 – Square: $25 + b^2 = 169$

Step 4 – Subtract 25 (finding a LEG = SUBTRACT): $b^2 = 144$

✓ Step 5 – Square root: $b = \sqrt{144} = 12$

ERROR ANALYSIS A student finds a missing leg when $c = 13$, $b = 5$ by writing

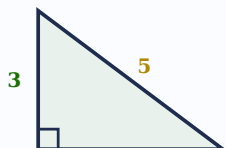
$a^2 = 13^2 + 5^2 = 194$. What's wrong?

To find a leg you SUBTRACT: $a^2 = c^2 - b^2 = 169 - 25 = 144 \rightarrow a = 12$. Adding is only for the hypotenuse.

4 Converse of the Pythagorean Theorem

✦ **THE CONVERSE:** Given three side lengths $a \leq b \leq c$, test: does $a^2 + b^2 = c^2$? If YES → right triangle. If NO → not a right triangle. Always test the two SMALLER sides against the LARGEST.

DOES $a^2 + b^2 = c^2$?



$$3^2 + 4^2 = 25 = 5^2 \checkmark$$

RIGHT triangle



$$2^2 + 3^2 = 13 \neq 4^2 = 16 \times$$

NOT a right triangle

MC Sides 3, 4, 5 — right triangle?

✓ $3^2 + 4^2 = 9 + 16 = 25$; $5^2 = 25$; $25 = 25 \rightarrow$ YES, right triangle.

MC Sides 2, 3, 4 — right triangle?

✓ $2^2 + 3^2 = 4 + 9 = 13$; $4^2 = 16$; $13 \neq 16 \rightarrow$ NO, not a right triangle.

MC Sides 5, 12, 13 — right triangle?

✓ $5^2 + 12^2 = 25 + 144 = 169$; $13^2 = 169$; $169 = 169 \rightarrow$ YES, right triangle.

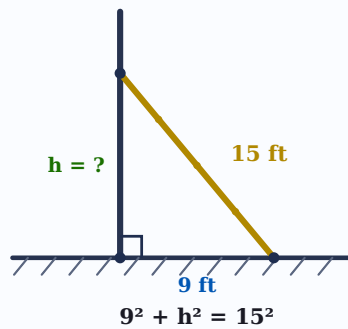
MC Sides 6, 7, 10 — right triangle?

✓ $6^2 + 7^2 = 36 + 49 = 85$; $10^2 = 100$; $85 \neq 100 \rightarrow$ NO, not a right triangle.

5 Word Problems

STRATEGY: (1) Read carefully · (2) Draw and label a diagram · (3) Identify legs vs hypotenuse · (4) Set up $a^2 + b^2 = c^2$ · (5) Solve step by step.

LADDER AGAINST A WALL



Problem — Ladder Against a Wall: A ladder 15 ft long leans against a wall. The base is 9 ft from the wall. How high does it reach?

Setup: $9^2 + h^2 = 15^2$

Square & subtract: $81 + h^2 = 225 \rightarrow h^2 = 144$

✓ **Square root:** $h = \sqrt{144} = 12 \text{ ft}$

6 Distance on the Coordinate Plane

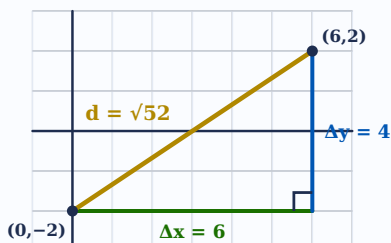
✦ **HOW IT WORKS:** Connect the two points to form a right triangle. The horizontal distance is $\Delta x = |x_2 - x_1|$ and the vertical distance is $\Delta y = |y_2 - y_1|$. Then use $a^2 + b^2 = c^2$ where $a = \Delta x$, $b = \Delta y$, and $c = \text{distance}$.

🔗 **NO DISTANCE FORMULA NEEDED!** You already know how to do this — just use the Pythagorean Theorem!

Example: Distance from (0, -2) to (6, 2)

Right triangle on the grid

DISTANCE = HYPOTENUSE



$$\Delta x: |6 - 0| = 6$$

$$\Delta y: |2 - (-2)| = 4$$

$$\text{Setup: } \Delta x^2 + \Delta y^2 = d^2 \rightarrow 36 + 16 = 52$$

$$\checkmark \text{ Answer: } d = \sqrt{52} \approx 7.21$$

⚠ **WATCH YOUR SIGNS:** When subtracting negative coordinates, remember: subtracting a negative means ADDING! Example: $5 - (-3) = 5 + 3 = 8$, NOT $5 - 3 = 2$.

🔧 PART 2 — Practice Problems

Show all work including the formula setup. Check on screen.

A Find the Hypotenuse

Formula: $c^2 = a^2 + b^2 \rightarrow c = \sqrt{a^2 + b^2}$.

1 Legs 3, 4. $c = ?$

C =

2 Legs 5, 12. $c = ?$

C =

3 Legs 6, 8. $c = ?$

C =

4 Legs 8, 15. $c = ?$

C =

5 Legs 7, 24. $c = ?$

C =

6 Legs 9, 12. $c = ?$

C =

7 Legs 9, 40. $c = ?$

C =

8 Legs 20, 21. $c = ?$

C =

B Find the Missing Leg

Formula: $a^2 = c^2 - b^2 \rightarrow a = \sqrt{c^2 - b^2}$ (SUBTRACT for a leg!).

1 Known leg 4, hyp 5. Leg = ?

leg =

2 Known leg 12, hyp 13. Leg = ?

leg =

3 Known leg 8, hyp 10. Leg = ?

leg =

4 Known leg 15, hyp 17. Leg = ?

leg =

5 Known leg 9, hyp 15. Leg = ?

leg =

6 Known leg 24, hyp 25. Leg = ?

leg =

7 Known leg 20, hyp 29. Leg = ?

leg =

8 Known leg 16, hyp 20. Leg = ?

leg =

C Word Problems

Strategy: Draw a picture! Label the sides. Decide: finding hypotenuse (ADD squares) or a leg (SUBTRACT squares)?

1. A 13-foot ladder leans against a wall. The base is 5 feet from the wall. How high does it reach?

$$h^2 + 5^2 = 13^2 \rightarrow h^2 = 169 - 25 = 144 \rightarrow \mathbf{h = 12 \text{ ft.}}$$

2. A rectangular TV is 36 inches wide and 15 inches tall. What is the diagonal length?

$$d^2 = 36^2 + 15^2 = 1296 + 225 = 1521 \rightarrow \mathbf{d = 39 \text{ in.}}$$

3. A baseball diamond is a square with 90-ft sides. Distance from home to second base?

$$d^2 = 90^2 + 90^2 = 16200 \rightarrow \mathbf{d \approx 127.3 \text{ ft.}}$$

4. A kite string is 100 m long. The kite is directly above a point 60 m away. How high is the kite?

$$h^2 + 60^2 = 100^2 \rightarrow h^2 = 10000 - 3600 = 6400 \rightarrow \mathbf{h = 80 \text{ m.}}$$

5. A support beam covers 9 ft horizontal and 12 ft vertical. How long is the beam?

$$c^2 = 9^2 + 12^2 = 81 + 144 = 225 \rightarrow \mathbf{c = 15 \text{ ft.}}$$

6. A pool is 24 m long and 10 m wide. A swimmer swims diagonally. How far?

$$d^2 = 24^2 + 10^2 = 576 + 100 = 676 \rightarrow \mathbf{d = 26 \text{ m.}}$$

D Distance on the Coordinate Plane

Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$ — it's just the Pythagorean Theorem on a grid!

1 $(1,2) \rightarrow (4,6)$. $d = ?$

d =

2 $(0,0) \rightarrow (3,4)$. $d = ?$

d =

3 $(-2,1) \rightarrow (4,9)$. $d = ?$

d =

4 $(1,-3) \rightarrow (7,5)$. $d = ?$

d =

5 $(-3,-1) \rightarrow (5,5)$. $d = ?$

d =

6 $(2,1) \rightarrow (5,5)$. $d = ?$

d =

★ PART 3 — Answer Key ★

Do not distribute to students until after the lesson.

★ Answers

Guided Notes

5.2 Example: $c = \sqrt{100} = 10$. Practice: 5, 10, 13, 17. 5.3 Example: $b^2 = 144 \rightarrow b = 12$. 5.4 Converse: (1) $25 = 25$ YES; (2) $13 \neq 16$ NO; (3) $169 = 169$ YES; (4) $85 \neq 100$ NO. 5.5 Ladder: $h^2 = 144 \rightarrow h = 12$ ft. 5.6 Distance: $\Delta x = 6, \Delta y = 4 \rightarrow 36 + 16 = 52 \rightarrow d = \sqrt{52} \approx 7.21$.

Part A — Hypotenuse

1) 5 2) 13 3) 10 4) 17 5) 25 6) 15 7) 41 8) 29.

Part B — Missing Leg

1) 3 2) 5 3) 6 4) 8 5) 12 6) 7 7) 21 8) 12.

Part C — Word Problems

1) $h^2 = 169 - 25 = 144 \rightarrow 12$ ft 2) $d^2 = 1296 + 225 = 1521 \rightarrow 39$ in 3) $d^2 = 16200 \rightarrow \approx 127.3$ ft 4) $h^2 = 10000 - 3600 = 6400 \rightarrow 80$ m 5) $c^2 = 225 \rightarrow 15$ ft 6) $d^2 = 676 \rightarrow 26$ m.

Part D — Distance

1) $\Delta x=3, \Delta y=4 \rightarrow 5$ 2) $3,4 \rightarrow 5$ 3) $6,8 \rightarrow 10$ 4) $6,8 \rightarrow 10$ 5) $8,6 \rightarrow 10$ 6) $3,4 \rightarrow 5$.

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